

**Département : Génie Électrique**  
**Master : ingénierie électrique et systèmes embarqués**

**TD : Ingénierie des circuits électriques**  
**Série 2**

**Exercise 1**

A 0.18- $\mu\text{m}$  fabrication process is specified to have  $t_{ox} = 4 \text{ nm}$ ,  $\mu_n = 450 \text{ cm}^2/\text{V.s}$ , and  $V_T = 0.5 \text{ V}$ .

- 1) Find the value of the process transconductance parameter  $k'_n$ .
- 2) For a MOSFET with minimum length fabricated in this process, find the required value of  $W$  so that the device exhibits a channel resistance  $r_{DS}$  of  $1 \text{ k}\Omega$  at  $v_{GS} = 1 \text{ V}$ .
- 3) Find the overdrive voltage  $V_{OV}$  required to operate a transistor having  $W/L = 20$  in saturation with  $I_D = 0.3 \text{ mA}$ .
- 4) What is the minimum value of  $V_{DS}$  needed?

**Exercise 2**

The PMOS transistor shown in Fig. 1 has  $V_{TP} = -1 \text{ V}$ ,  $k'_p = 60 \mu\text{A}/\text{V}^2$ , and  $W/L = 10$ .

- (a) Find the range of  $V_G$  for which the transistor conducts.
- (b) In terms of  $V_G$ , find the range of  $V_D$  for which the transistor operates in the triode region.
- (c) In terms of  $V_G$ , find the range of  $V_D$  for which the transistor operates in saturation.
- (d) Neglecting channel-length modulation ( $\lambda = 0$ ), find the values of  $|V_{OV}|$  and  $V_G$  and the corresponding range of  $V_D$  to operate the transistor of figure 1 in the saturation mode with  $I_D = 75 \mu\text{A}$ .

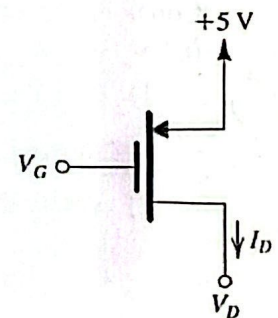


Figure 1

- (e) If  $\lambda = -0.02 \text{ V}^{-1}$ , find the value of  $r_o$  corresponding to the overdrive voltage determined in (d).
- (f) For  $\lambda = -0.02 \text{ V}^{-1}$  and for the value of  $V_{OV}$  determined in (d), find  $I_D$  at  $V_D = +3 \text{ V}$  and at  $V_D = 0 \text{ V}$ ; hence, calculate the value of the apparent output resistance in saturation. Compare to the value found in (e).

**Exercise 3**

Consider the circuit in Fig. 2, the MOSFET has  $V_{TN} = 0.5 \text{ V}$ ,  $\mu_n C_{ox} = 0.4 \text{ mA}/\text{V}^2$ ,  $W/L = 0.72 \mu\text{m}/0.18 \mu\text{m}$ , and  $\lambda = 0$ .

- 1) Find the value of  $R$  that results in  $V_D = 0.7 \text{ V}$ .



Figure 3 shows a circuit obtained by augmenting the circuit of Fig. 2 with a transistor  $Q_2$  identical to  $Q_1$  and a resistance  $R_2$ .

- Find the value of  $R_2$  that results in  $Q_2$  operating at the edge of the saturation region.

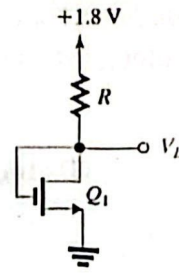


Figure 2

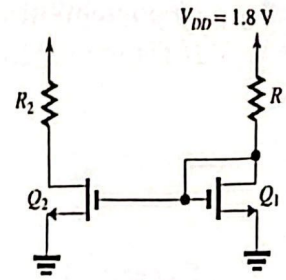


Figure 3

#### Exercise 4

For the configurations shown in Fig. 4, determine the small-signal resistances  $R_X$  and  $R_Y$ . Assume  $\lambda \neq 0$ .

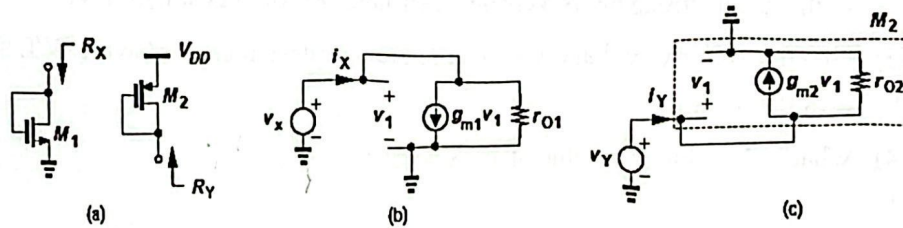


Figure 4

#### Exercise 5

Consider the circuit of figure 5, with  $\mu nC_{ox} = 100 \mu A/V^2$ ,  $V_{th} = 0.5 V$ , and  $\lambda = 0$ .

- Calculate the small-signal voltage gain of the CS stage shown in Fig. 5 if  $I_D = 1 mA$ .
- Verify that  $M_1$  operates in saturation.

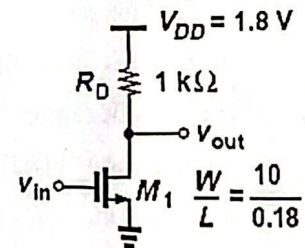


Figure 5

#### Exercise 6

Consider the amplifier of Fig. 6(a) with  $V_{DD} = 1.8 V$ ,  $R_D = 17.5 k\Omega$ , and with a MOSFET specified to have  $V_t = 0.4 V$ ,  $kn = 4 mA/V^2$ , and  $\lambda = 0$ .

- Determine the coordinates of the end points of the active-region segment of the VTC.
- Determine  $V_{DS|C}$  assuming  $V_{GS|C} = V_{DD}$ .

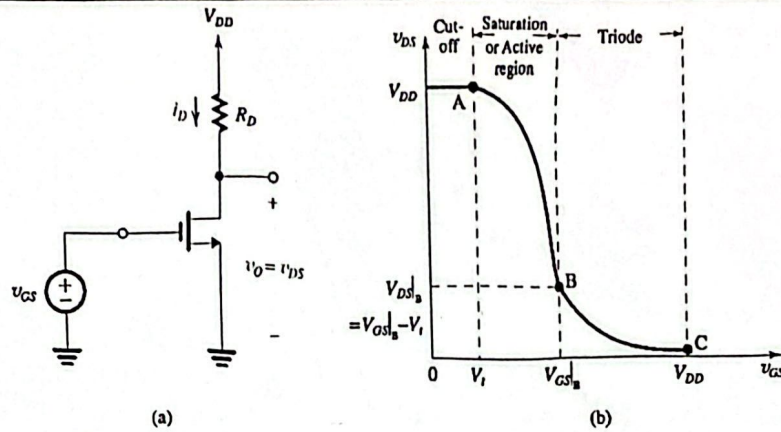


Figure 6 (a) An NMOS amplifier and (b) its Voltage transfer characteristic

### Exercise 7

Figure 7 shows a discrete MOSFET amplifier utilizing a drain-to-gate resistance  $R_G$  for biasing purposes. The transistor has  $V_{tn} = 1.5\text{V}$ ,  $k'_n (W/L) = 0.25 \text{ mA/V}^2$ , and  $V_A = 50 \text{ V}$ . Assume the coupling capacitors to be sufficiently large so as to act as short circuits at the signal frequencies of interest.

- 1) determine its small-signal voltage gain,
- 2) determine the input and the output resistances,
- 3) determine the largest allowable input signal with keeping transistor NMOS in saturation region.

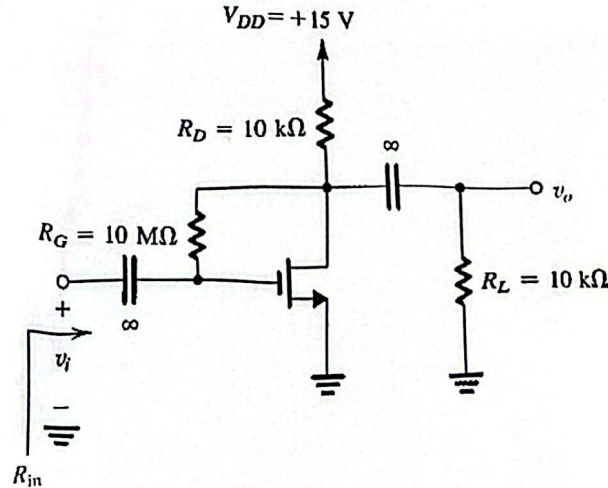


Figure 7 amplifier circuit



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**Série 3**

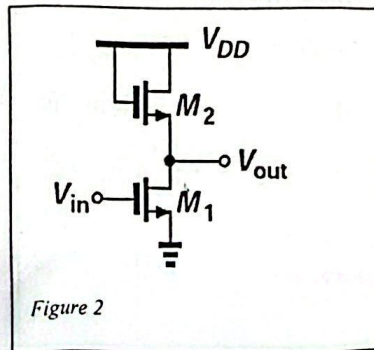
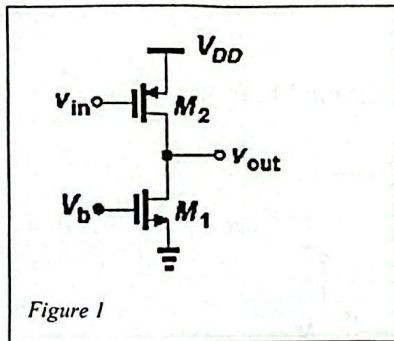
**Exercise 1**

Figure 1 shows a PMOS CS stage using an NMOS current source load.

- 1) Compute the output resistance of the circuit in Fig.1
- 2) Compute the voltage gain of the circuit.

Consider the CS stage given in Fig.2, with channel-length modulation not neglected

- 3) Compute the output resistance and the voltage gain of the circuit in Fig 2

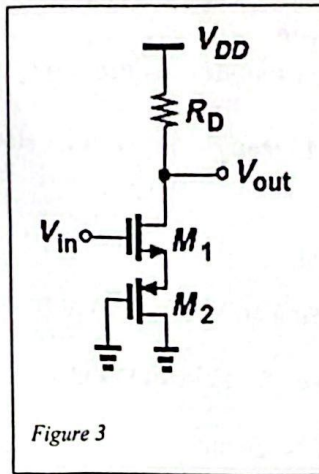


**Exercise 2**

- 1) Compute the voltage gain of the CS degenerated circuit shown in Fig. 3 if  $\lambda = 0$ , and if  $\lambda \neq 0$ ?

To design the integrated circuit of CS in fig3 with an active component, we can replace resistance  $R_D$  by diode connected PMOS.

- 2) Give the integrated circuit schematic?
- 3) Compute the voltage gain if  $\lambda=0$ ?



### Exercise 3

- 1) Compute the output resistance  $R_{out}$  of the circuit in Fig. 4 if  $M1$  and  $M2$  are identical,  $M1$  in saturation region
- 2) Compute the  $R_{out}$  of the circuit in Fig 5. assume  $M1$  and  $M2$  in saturation.

